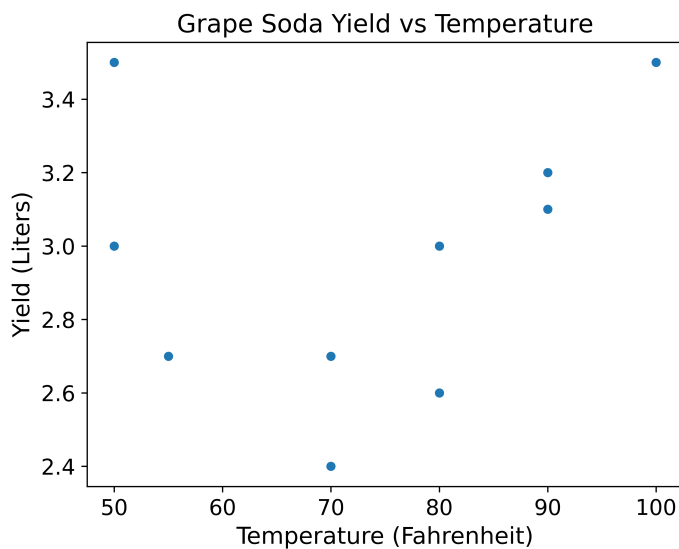


Discussion #5 Solutions

Note: Your TA will probably not cover all the problems. This is totally fine; the discussion worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures, labs, and homework.

Simple Linear Regression

1. Lillian and Prabhleen were watching their favorite chemistry Youtuber [NileRed](#) experimenting with turning gloves into grape soda and wanted to try it themselves. The experiment was done at various temperatures and yielded various amounts of grape soda. Since this reaction is very costly, they were only able to do it 10 times. This data set of size $n = 10$ (Yield vs Temperature data) contains measurements of yield from an experiment done at six different temperature levels. The variables are $y = \text{yield in liters}$ and $x = \text{temperature in degrees Fahrenheit}$. Below is a scatter plot of our data.

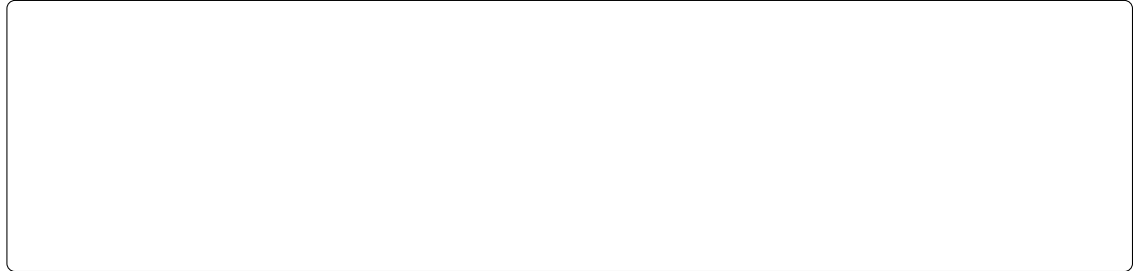


σ_x	σ_y	r	$\bar{x}$	$\bar{y}$
15	0.3	0.50	75.00	3

Note: The values of σ_y and σ_x are slightly different from what the table above shows, this was modified to make calculations slightly easier during discussion.

- (a) Given the above statistics, **calculate** the slope ($\hat{\theta}_1$) and y-intercept ($\hat{\theta}_0$) of the line of best fit using Mean Squared Error (MSE) as our loss function and **plot the line on the graph above**:

$$y = \hat{\theta}_0 + \hat{\theta}_1 x$$

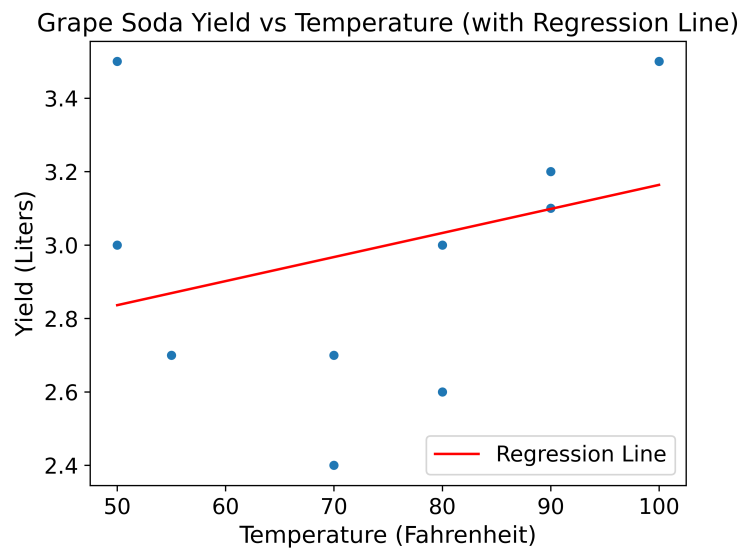


Solution: $\hat{\theta}_1 = r \frac{\sigma_y}{\sigma_x}$

$$\hat{\theta}_1 = 0.5 \frac{0.3}{15} = 0.01$$

$$\hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x}$$

$$\hat{\theta}_0 = 3 - 75.00 * 0.01 = 3 - 0.75 = 2.25$$



Note: The intercept and slope of the line in the plot above do not match up exactly with the values we calculate.

- (b) Which of the following relations most closely represent the relationship we see between Temperature (x) and Yield (y)?
- ☐ A. $y = \theta_2 x^2$
 - ☒ B. $y = \theta_2 x^2 + \theta_1 x + \theta_0$
 - ☐ C. $y = \theta_1 \log x + \theta_0$
 - ☐ D. $y = \theta_1 x + \theta_0$

Solution: Based on the shape of the original and residual plots, we can see that the graph follows a curved, or quadratic (x^2), pattern as opposed to a linear one (x) or a logarithmic one ($\log x$). To choose between choice A or B, we can look at the intercept of the original graph; because all values have a yield above 2.4, we expect the intercept θ_0 to be a nonzero value, so we choose B instead of A.